The Answers

1. Basic principles of semiconductors

a) Different physical parameters influence the conductivity of a material.

i) the density of carriers and the scattering events (influence velocity/mobility). [2]

ii) it increases [2]

iii) because the carrier density is increasing faster than the scattering events are decreasing the mobility. [4]

iv) it decreases [2]

v) because the carrier density remains approximately constant but the scattering events are increasing thus reducing the velocity of the carriers. [4]

b) Different carrier transport processes determine the currents through a device.

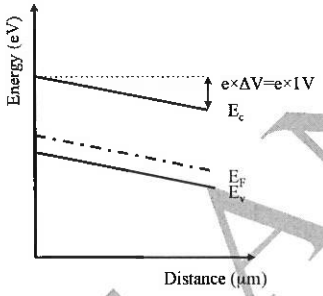
i) electron drift. [2]

ii) electron diffusion and hole drift in the charge neutral p-type region

hole diffusion and electron drift in the charge neutral n-type region

across the depletion region: diffusion of electrons and drift of holes from n to p region, diffusion of holes and drift of electrons from the p to n region. [6]

c) Energy band diagrams.

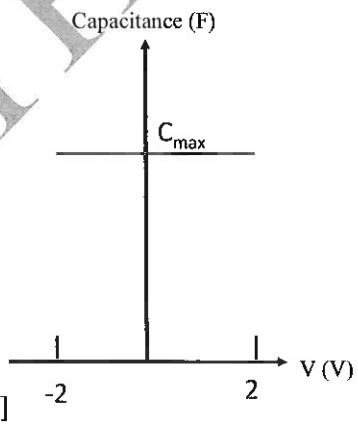


i) [4]

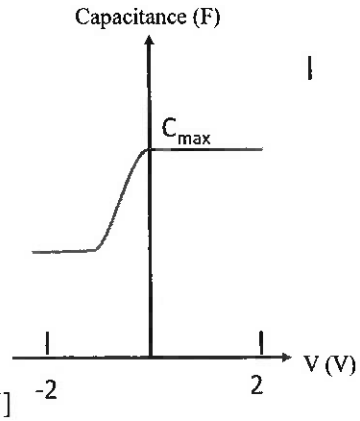
ii) 1 eV [2]

iii) 1 eV. [2]

d) $C_{max} = \frac{4\times8.85\times10^{-12}\ \mathrm{F/m}}{100\times10^{-9}m}100\times10^{-12} = 3.54\times10^{-14}F$ for both cases



i) [5]



ii) [5]

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2. MOSFETs

a) The threshold voltage is the gate voltage that needs to be applied to generate an inversion layer carrier density at the Si/SiO_2 interface of the same magnitude as the carrier density in the bulk. Inversion charge has the opposite charge sign compared to the bulk charge. [4]

b) The energy band diagram from gate through the oxide into the silicon (MOS).

i) p-type. The Fermi level in Si lies closer to valence band than to the conduction band or equivalently the Fermi level lies below the intrinsic level (midgap) in Si. [2]

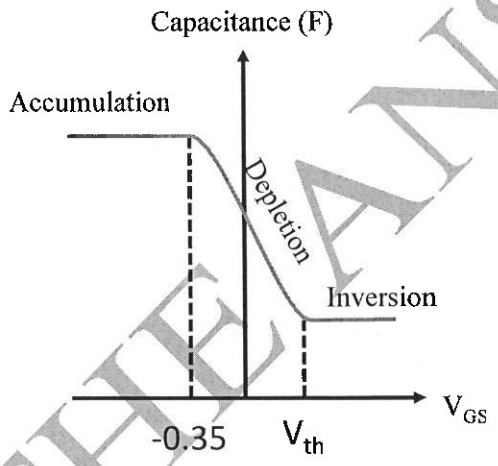
ii) zero. Because the Fermi levels are constant across the structure. [2]

iii) $V_{FB} = V_{ox} + V_s = 0.1\ eV + 0.25\ eV = 0.35\ eV$. [4]

iv) The workfunction of the metal is smaller than the workfunction of the Si. Because when the free energy of electrons in the metal is higher than that in Si before contact, the free energy of electrons in the Si must increase upon contact and thus E_c comes closer to E_F as then free electrons can be found at higher energy level. [4]

v) The threshold voltage will be lower than the case presented in fig. 2.1. For the same initial band bending we need less further bending to ensure the distance between E_c and E_F at the Si/SiO_2 interface is equal to the distance between E_F and E_v in the bulk as E_F-E_v in bulk has increased with lower N_A. [4]

c) [10]

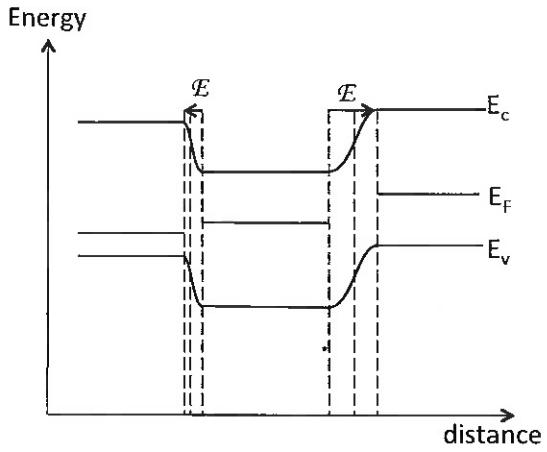


Marks distribution:
Right orientation: 2
Right naming of areas: 3
Shift towards negative V_{GS}: 3
Values on voltage axis: 2

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3. BJT

a) [10]



Marks distribution:
Correct doping: 3
Correct flat PE: 1
Correct movement of E_F with bias: 2
Correct relative dimensions of depletion widths: 2
Correct position and direction of $\mathcal{E}$: 2

b)

i) the emitter current density. [3]

$$J_E = eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right) - n_{p_0}}{X_E} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right) - p_{n_0} exp\left(V_{CB}/V_T\right)}{W_B}$$

$$V_{CB} < 0V \;\&\; exp\left(V_{EB}/V_T\right) \gg 1$$

$$J_E \approx eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right)}{X_E} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B}$$

$$J_E \approx e \times n_i^2 \times exp\left(V_{EB}/V_T\right)\left(\frac{D_n}{N_A X_E} + \frac{D_p}{N_D W_B}\right) =$$
$$1.6\,10^{-19}\,(1.45\,10^{10})^2 exp\left(\frac{0.5}{0.26}\right)\left(\frac{20}{10^{18}\times 10^{-4}} + \frac{12}{10^{16}\times 0.8\times 10^{-4}}\right) = 4.6\times 10^{-11} +$$
$$3.5\times 10^{-9} \approx 3.5\times 10^{-9} A/cm^2$$

ii) the collector current density. [3]

$$J_C = eD_n \frac{n_{p_0} exp\left(V_{CB}/V_T\right) - n_{p_0}}{X_C} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right) - p_{n_0} exp\left(V_{CB}/V_T\right)}{W_B}$$

$$V_{CB} < 0V$$

$$J_C \approx eD_n \frac{-n_{p_0}}{X_C} + eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B}$$

$$exp\left(V_{EB}/V_T\right) \gg 1 \;\&\; W_B \ll X_C$$

$$J_C \approx eD_p \frac{p_{n_0} exp\left(V_{EB}/V_T\right)}{W_B} = 3.5\times 10^{-9} A/cm^2$$

iii) the base current density. [2]

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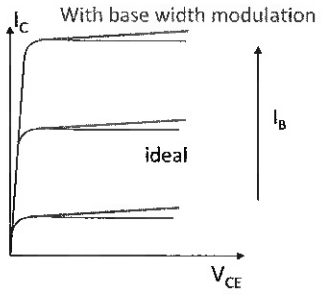
$$J_B = J_E - J_C \approx eD_n \frac{n_{p_0} exp\left(V_{EB}/V_T\right)}{X_E} = 4.6\times10^{-11} A/cm^2$$

iv) the current gain. [2]

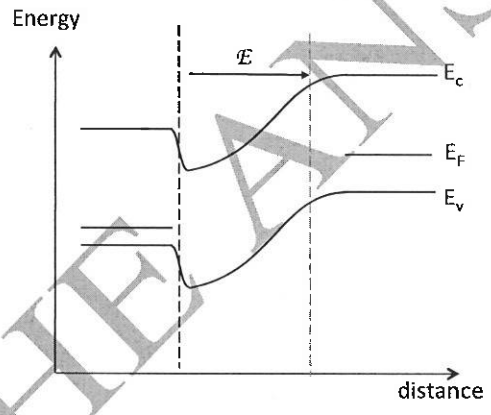
$\beta = \frac{J_C}{J_B} = \frac{D_p X_E N_A}{D_n W_B N_D} = \frac{3.5\times10^{-9}}{4.6\times10^{-11}} = 75$ with N_A the doping in the emitter and N_D the doping in the base

c) Output characteristics.

i) When taking the depletion region width that extend into the base into account then the output characteristics show a slope for higher V_{CE} values due to base width modulation – the effective width of the base decreases with increasing reverse bias across the base-collector junction (thus V_{CE} increasing). I_C is inversely proportional to base width. Output conductance $g_0 > 0$ S. [3]



ii) . [3]



iii) When the base is completely depleted then there is a strong electric field and the energy band diagram resembles that of a diode (doped-depletion-doped). We can expect exponential currents when V_{BC} lowers the EB potential barrier for V_{CE} increasing. [2]

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